MemGUI-Agent: An End-to-End Long-Horizon Mobile GUI Agent with Proactive Context Management

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Paper version: arXiv 2606.19926v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *passively accumulates per-step records: prompt explosion and dilution of critical cross-app facts*

ReAct-style logs grow with the horizon; prices, ids and copied text get diluted across app switches.

Why long-horizon mobile tasks fail

1. **Prompt explosion.** History grows with steps.
2. **Information loss.** Cross-app facts diluted or truncated.
3. **Not in the policy.** External memory is outside.

Evaluation setup

MemGUI-Bench · 128 | MobileWorld · 117 | 26 Android apps | Qwen3-VL 8B / 235B | budget 2.5g + 1 | 3 attempts (Bench) | 50 steps (MW), 1 try

- **All headline results online**; offline metrics kept apart.
- **Held out:** MobileWorld apps and protocol differ; zero task overlap.

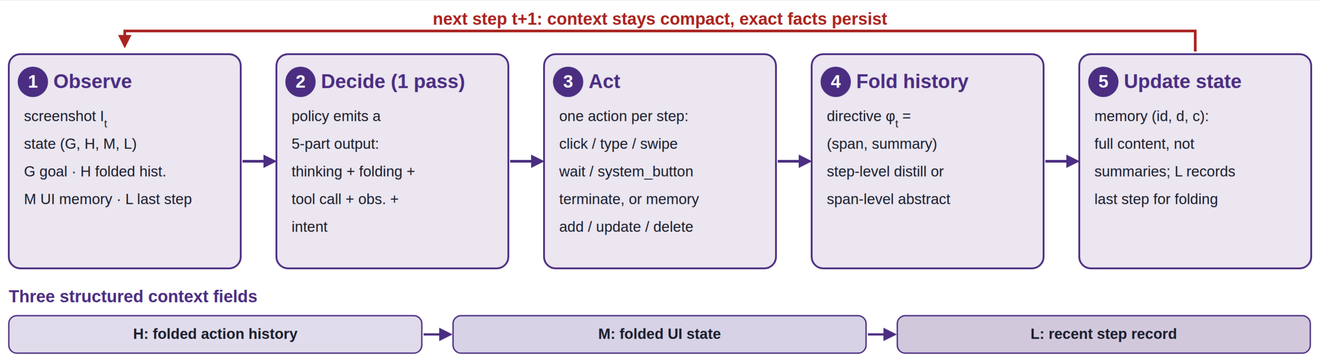
Our read (CUA Radar)

62	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval ×2; held-out env; same-backbone deltas; ablation; cost. **Not:** run variance; validator; code announced.
- **Flags:** zero-shot ConAct degrades 2B–235B-Instruct (AP15); folding is lossy (AP18).
- Pattern C16. Authors' numbers, not rerun.

Method: ConAct — context management as first-class actions

- **Sees / acts:** screenshot + state (G, H, M, L) in; UI action or memory write out, one per step.
- **One forward pass** emits thinking, folding, tool call, UI observation, action intent.
- **Signal:** SFT on MemGUI-3K — 2,956 teacher trajectories, reasonable steps only.



Our redraw (not the paper's figure): the ConAct step loop over the three structured context fields.

$$y_t = (\tau_t,\ \varphi_t,\ a_t,\ o_t,\ \iota_t) \sim \pi_\theta(\cdot \mid I_t, S_t)$$

one pass emits reasoning, folding, action, observation, intent

Key results: MemGUI-Bench Pass@1 (%)

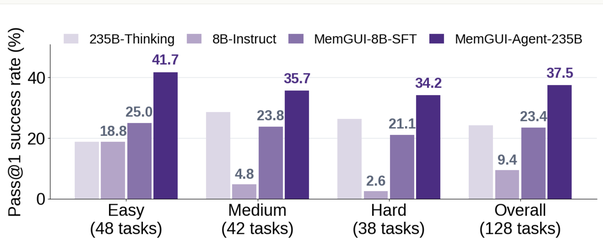


Table 2, redrawn: online runs, up to 3 attempts, budget 2.5g + 1, Qwen3-VL backbones.

+13.3 P@1 / +15.6 P@3 over the same 235B backbone

SO WHAT → *All Qwen3-VL backbones; best Gemini-2.5-Pro framework (M3A) 32.8 / 47.7. IRR 46.8 vs 30.0 same backbone.*

Ablation: ConAct components

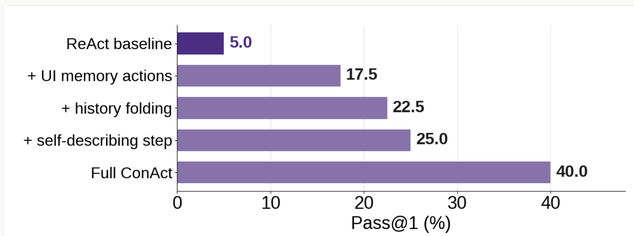


Table 5, redrawn: MemGUI-Bench-40, zero-shot Qwen3-VL-235B-Thinking.

SO WHAT → *No single part suffices; full ConAct 40.0 P@1 / 62.5 P@3 vs ReAct 5.0 / 27.5. Memory actions alone triple P@1 (5.0 → 17.5); folding alone hits P@3 32.5 vs 42.5 for memory; IRR 19.5 → 51.0. Offline, deep-fold ratio 26.1 vs 23.2 gold. 8B-SFT gains concentrate on Hard (+18.5) vs Easy (+6.2); errors fall 99 → 58; format tags 99.9.*

Headline Numbers

62.5%
MemGUI-Bench Pass@3
zero-shot 235B; best 47.7%

- **37.5%** Pass@1, same bench
- **29.1%** MobileWorld 235B SR
- **17.9%** MobileWorld 8B SR

The paper's own controls

CONTROL	RESULT	VS
Scale sweep	only 235B-Thinking	2B 10.0→5.0; 8B 12.5→7.5
Component ablation	folding alone 22.5 P@1	full ConAct 40.0 P@1
Held-out transfer	MobileWorld 29.1%	17.9% 8B-SFT, +8.5
Offline skill check	trigger F1 19.9 → 48.0	gold-context steps

Takeaway

Context management as policy actions: zero-shot ConAct 62.5% Pass@3; SFT lifts the 8B model +14.0 Pass@1 and transfers to MobileWorld.

2,956 trajs · 26 apps | avg 28.8 steps/traj | 65.1% use memory

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